

DA HACKATHON

Dark Store Down- Quick Commerce Analytics Challenge

The Challenge

Imagine you are part of the **Data & Analytics team of a quick-commerce company**.

The company operates multiple dark stores and handles thousands of customer orders across different locations.

The business team has shared a set of operational data with you — covering areas such as **stores, orders, deliveries, products and promotions**.

Your job is simple to understand, but not necessarily simple to solve:

Explore the data, identify meaningful business problems or opportunities, and use data to explain what is happening and what the business should think about next.

There is **no single prescribed approach**.

The expectation is not just to create charts.

You need to think like an analyst who has been given a real business problem with incomplete information.

Your Mission

As a team, investigate the business using the data provided and build a story around the areas you find interesting or important.

You may discover problems related to:

- Store performance
- Orders & revenue
- Delivery & fulfilment
- Products
- Promotions & discounts
- Customer behaviour
- Operational efficiency
- Profitability

- Data quality
- Or something else that the data allows you to investigate

These are only examples — you are not required to analyse all of them.

The interesting part of the hackathon is deciding:

What is worth investigating?

and then proving your thinking using data.

Your Team

Each team will have **at least 3 members**.

How you divide the work is completely up to your team.

You may:

- divide the analysis by business area,
- divide work by tools,
- have one person focus on data preparation,
- have another focus on analysis,
- have another focus on storytelling and presentation,

or use any other approach that works for your team.

There is **no fixed role structure**.

The important thing is that everyone contributes and understands the final work.

The Data

You will receive a collection of datasets covering different parts of the business.

The resources include data related to:

1. Store Performance

Information related to dark stores and their operating details.

2. Delivery & Fulfilment

Information related to delivery partners and delivery/trip activity.

3. Products & Promotions

Information related to products, order items and promotional codes.

4. Shared Order Data

A common order-level dataset that can be used across different analyses.

The data has intentionally been provided in different files.

Part of the challenge is to **understand the data, identify relationships between datasets, clean what needs to be cleaned, and decide how the data can be used effectively.**

The original dataset contains order, delivery and product-related information across multiple stores and a period of several weeks.

What You Need to Build

Every team must create **at least 2–3 dashboards/reports/analytical outputs** covering **different topics or business questions.**

For example, your team might decide to investigate:

Store performance + Delivery efficiency + Promotions

or

Product performance + Customer orders + Operational efficiency

or something completely different.

The choice is yours.

Your outputs can be:

- Excel dashboards
- Power BI dashboards
- SQL-based analysis
- Python analysis
- Analytical reports
- A combination of these
- Or even another approach/tool if you believe it helps solve your problem

There is no tool restriction.

You can complete the entire hackathon using:

Excel only

or

SQL only

or

Python only

or

Excel + SQL + Python + Power BI

or any other combination.

We care much more about the quality of your thinking and analysis than the tool you use.

Think Beyond the Obvious

Do not approach this as:

"What charts can I make from this dataset?"

Instead, think:

"If I were working for this company, what would I want to understand?"

You may start with something simple and then discover something more interesting.

For example:

Observation → Question → Investigation → Insight → Business implication

A good analysis may begin with a simple observation but should try to understand **why** something is happening.

You are encouraged to go deeper whenever the data allows it.

Your Analysis Should Have a Story

By the end of the hackathon, your team should be able to answer:

What did we investigate?

What business question or problem did your team choose?

What did we find?

What does the data tell you?

Why does it matter?

What could this mean for the business?

What should the business think about?

What action, decision, experiment or further investigation would you suggest?

You do **not** need to have a perfect answer.

A well-supported finding with a thoughtful recommendation is much more valuable than a dashboard full of random charts.

Quality > Quantity

Creating 20 charts does not automatically make your analysis better.

A strong dashboard/report might contain only a few carefully selected visuals but answer an important question clearly.

We will look at things such as:

- Is the business question meaningful?
- Is the analysis relevant to that question?
- Did you investigate beyond the obvious?
- Are the numbers correct?
- Did you consider the quality of the data?
- Are your conclusions actually supported by the data?
- Can you explain why your finding matters?
- Is the output easy to understand?
- Does the final presentation tell a clear story?

The original hackathon material also emphasises analytical correctness, business insight, communication/storytelling, cleaning rigour and quality of the final deliverable.

Data Cleaning Is Part of the Challenge

Do not assume that every dataset is immediately analysis-ready.

Some of the provided datasets may require you to:

- understand inconsistent values,
- handle missing data,
- identify duplicates,
- standardise fields,
- understand data types,
- validate relationships,
- or make other cleaning decisions.

You should be able to explain:

What did we change, why did we change it, and how might it affect our analysis?

The provided hackathon data deliberately includes files where cleaning is part of the analytical work.

You Are NOT Given a Step-by-Step Solution

This is important.

You will **not** be given:

- a list of questions to solve,
- a fixed list of KPIs,
- a fixed dashboard layout,
- a fixed SQL query,
- a fixed Python analysis,
- a list of charts to create,
- or a step-by-step procedure.

That is intentional.

This is a **hackathon**, not a regular module project.

You are expected to explore, discuss, experiment and make decisions as a team.

There may be multiple valid ways of approaching the same business problem.

Use Your Resources

You are allowed — and encouraged — to use the resources available to you.

You can use:

- Google
- Documentation
- YouTube
- AI tools
- ChatGPT / Gemini / Claude
- Your course material
- Your teammates
- Mentors and instructors

But remember:

AI can help you analyse the problem. It cannot replace your understanding of the analysis.

If an AI tool gives you a query, formula, code or insight, make sure your team understands what it is doing.

During the presentation, you may be asked:

"Why did you calculate this?"

"Why did you choose this metric?"

"How did you arrive at this conclusion?"

"What does this number actually mean?"

Be prepared to defend your work.

What We Will Evaluate

At the end of the hackathon, we will primarily look at four things:

1. Quality of Analysis

How deeply and clearly did you investigate the business question you selected?

Did you stop at the first observation, or did you investigate further?

2. Quality of Insights

Are your insights meaningful and supported by data?

Can you explain the **why**, not just the **what**?

3. Quality of Output

Are your dashboards/reports:

- clear,
- organised,
- visually understandable,
- relevant,
- and easy for someone else to interpret?

4. Quality of Presentation

Can your team clearly communicate:

- the problem,
- your approach,
- your findings,
- the evidence,
- and your recommendations?

The presentation is not simply about showing your dashboard.

It is about telling the **story behind your analysis**.

One Important Rule

Your team must create **at least 2–3 meaningful analytical outputs covering different topics/business questions.**

Do not simply create three versions of the same dashboard.

The purpose is to demonstrate that your team can explore the business from different perspectives.

There Is No "Perfect" Answer

You may discover something that another team does not.

Another team may approach the same data completely differently.

That is okay.

We are not expecting every team to produce the same analysis.

What matters is whether you can:

Find → Investigate → Validate → Explain → Communicate

Final Presentation

Your team will present its work to the judges.

Your presentation should help the audience understand:

What did you investigate?

↓

Why did you investigate it?

↓

What did you discover?

↓

What evidence supports your finding?

↓

What does it mean for the business?

↓

What would you recommend / investigate next?

Keep the audience in mind.

A technically impressive analysis that nobody understands is not a strong business presentation.

Final Reminder

Guys, this is a hackathon.

It is not a normal module project where a mentor will tell you:

"First do this query, then create this chart, then calculate this KPI."

You have the data.

You have the tools.

You have your team.

Now **think**.

Use your common sense.

Ask questions.

Explore the data.

Use AI.

Try things.

Challenge each other's assumptions.

And most importantly — **take ownership of the problem**.

If you have a specific doubt or get genuinely stuck, you can always reach out to the doubt team or instructor.

But don't wait for someone to tell you exactly what to do.



The problem is intentionally open-ended.

Your team's job is to decide what is worth solving.